

32-bit microcontrollers

BGR module

Applicable to:

Series	Product Model	Series	Product Model
HC32L13	HC32L130E8PA	HC32F03	HC32F030E8PA
	HC32L130F8UA		HC32F030F8UA
	HC32L130J8TA		HC32F030F8TA
	HC32L136J8TA		HC32F030H8TA
	HC32L136K8TA		HC32F030J8TA
			HC32F030K8TA
HC32L07	HC32L072PATA	HC32F07	HC32F072PATA
	HC32L072KATA		HC32F072KATA
	HC32L072JATA		HC32F072JATA
	HC32L073PATA		
	HC32L073KATA		
	HC32L073JATA		
HC32L17	HC32L176PATA	HC32F17	HC32F176PATA
	HC32L176MATA		HC32F176MATA
	HC32L176KATA		HC32F176KATA
	HC32L176JATA		HC32F176JATA
	HC32L170JATA		HC32F170JATA
	HC32L170FAUA		HC32F170FAUA
HC32L19	HC32L196PCTA	HC32F19	HC32F196PCTA
	HC32L196MCTA		HC32F196MCTA
	HC32L196KCTA		HC32F196KCTA
	HC32L196JCTA		HC32F196JCTA
	HC32L190JCTA		HC32F190JCTA
	HC32L190FCUA		HC32F190FCUA

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1. Abstract

This application note primarily introduces the use of the BGR module in Huada Semiconductor MCUs*.

This application note includes:

- BGR Module
- BGR in PLL
- BGR in VC
- BGR in ADC
- BGR in OPA

Note:

- This application note is supplementary material for Huada Semiconductor MCUs* and cannot replace the user manual. For specific functions and register operations, please refer to the user manual.

2. Function Introduction

This article explains the application and usage precautions of the BGR module in Huada Semiconductor's MCU*.

**Supported models are listed on the cover.*

3. BGR Module

3.1 BGR Usage

- After starting BGR, wait 20 μ s for it to output a high-precision reference voltage.
- When using ADC, VC, OPA, or PLL, BGR must be enabled first and allowed to stabilize (wait 20 μ s). Refer to the relevant analog module chapters for detailed settings.
- ADC external clock must be enabled when configuring BGR; otherwise, the BGR register cannot be written to.

3.2 BGR in PLL

- The BGR is automatically enabled when the PLL is turned on, requiring no manual configuration.

3.3 BGR Usage in VC

- Using VC requires enabling the BGR.
 - Enable the ADC peripheral clock;
 - Set BGR_CR.BGR_EN to 1 to enable the BGR module;
 - Delay for 20 μ s to wait for the BGR module to complete startup.
- The BGR can output 1.2V as the N-terminal input of VC_x (x = 0, 1).

3.4 BGR Usage in ADC

- Using the ADC requires enabling the BGR.
 - Enable the ADC peripheral clock;
 - Set BGR_CR.BGR_EN to 1 to enable the BGR module;
 - Set ADC_CR0.En to 1 to enable the ADC module;
 - Delay for 20 μ s to wait for the ADC and BGR modules to complete startup.

3.5 BGR Usage in OPA

- Using OPA requires enabling BGR.
 - Enable the ADC peripheral clock;
 - Set BGR_CR.BGR_EN to 1 to enable the BGR module;
 - Delay for 20μs to wait for the BGR module to complete startup.

4. Reference Examples and Drivers

Through the above introduction, and in conjunction with this series of user manuals, we have gained a deeper understanding of the functions and operation methods of the BGR module in Huada Semiconductor's MCU*.

Huada Semiconductor (HDSC) also provides a driver library for this module. Users can further familiarize themselves with the module and the application of the driver library by opening sample projects. In actual development, users can also directly refer to the samples and use the driver library to quickly implement operations on this module.

5. Summary

The preceding sections briefly introduced the BGR module of Huada Semiconductor's MCU*, and explained its usage in detail. For a more in-depth understanding of the module's usage and operation during actual application development, users should refer to the corresponding user manual. The driver library mentioned in this article can be used for further experimentation and learning, or directly applied in actual development.

6. Other Information

Technical support information: www.hdsc.com.cn

**Supported models are listed on the cover.*

7. Version Information & Contact Information

Date	Version	Change History
2018/6/20	Rev1.0	Initial release.
2018/9/3	Rev1.1	Updated supported product models.
2019/6/24	Rev1.2	Updated supported product models.



If you have any comments or suggestions during the purchase and use process, please feel free to contact us.

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